

MTH302 Business Mathematics & Statistics

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Tuesday, July 12, 2011

10 Solved Quizzes

All Quizzes have been solved in Conference of MTH302

Arranged By ●*SHINE angel*● And *Vu Visioner*

Question # 1 of 10 (Start time: 06:25:48 PM) Total Marks: 1

Which of the following formula is correct?

Select correct option:

=BINOMDIST(7, 4, 0.5, FALSE)

=BINOMDIST(-4, 7, 1.5, FALSE)

=BINOMDIST(4, -7, 1.5, FALSE)

=BINOMDIST(4, 7, 0.5, FALSE)

Question # 2 of 10 (Start time: 06:27:25 PM) Total Marks: 1

Attendance in a factory shows 7 absences. What is probability that on a given day there will be more than 8 people absent?

Select correct option:

0.2709

0.184

0.5

None of these.

Question # 3 of 10 (Start time: 06:28:59 PM) Total Marks: 1

Formulation for Poisson distribution i-e "POISSON(x,mean,cumulative)" is truncated if 'x' is

Select correct option:

Non-integer

Integer

Integer but expected numeric value

Integer but non-expected value

Question # 4 of 10 (Start time: 06:30:04 PM) Total Marks: 1

In Binomial distribution, the outcome on one trial the outcome on other trials.

Select correct option:

Does not affect

Does affect

May affect

None of these

Question # 5 of 10 (Start time: 06:30:55 PM) Total Marks: 1

The mean of a Poisson distribution is 5 while its standard deviation is 4. then what option is correct about this statement.

Select correct option:

In some cases this result is possible.

Mean should be always less than standard deviation .

Mean and standard deviation are always equal.

Only option 1 & 3.

Question # 6 of 10 (Start time: 06:32:02 PM) Total Marks: 1

In binomial distribution, the experiment is repeated for

Select correct option:

Variable number of times

Fixed number of times

Many number of times

None of these

Question # 7 of 10 (Start time: 06:33:05 PM) Total Marks: 1

Which of the following is NOT applicable to a Poisson Distribution?

Select correct option:

It is used to compute the probability of rare events.

Every event is independent of every other event.

The range for the number of events that could occur is 0,1,2,3, ...

Every event is dependent of every other event.

Question # 8 of 10 (Start time: 06:34:38 PM) Total Marks: 1

The successive binomial trials are assumed to be

Select correct option:

Independent

Dependent

May be independent

None of these

Question # 9 of 10 (Start time: 06:35:40 PM) Total Marks: 1

Probability distribution is a set of possible values that a random variable can assume and their associated probabilities of occurrence.

Select correct option:

True

False

Question # 10 of 10 (Start time: 06:36:31 PM) Total Marks: 1

In negative binomial distribution, the probability of success, p

Select correct option:

Remains the same on every trial

Remains constant for different outcomes

Change on every trial

None of these

Question # 1 of 10 (Start time: 06:39:20 PM) Total Marks: 1

The sum of all probabilities for any poisson density function is always ____ .

Select correct option:

0

1

2

None of these.

Question # 2 of 10 (Start time: 06:40:17 PM) Total Marks: 1

The number of telephone calls that pass through a switchboard has a Poisson distribution with mean equal to 2 per minute. The probability that no telephone calls pass through the switch board in two consecutive minutes is:

Select correct option:

0.2707

0.0517

0.0183

0.1353

: Question # 3 of 10 (Start time: 06:41:27 PM) Total Marks: 1

If the mean & standard deviation on an examination are '74' & '12' respectively and the grades ' $x=74$ ' then the scores (z) in standard units of students receiving are

Select correct option:

0.8378

-0.8378

1

0

Question # 4 of 10 (Start time: 06:42:48 PM) Total Marks: 1

A pair of fair die is rolled. What will be the probability of "Sum of dots on upper sides is odd number?

Select correct option:

18/36

15/36

10/36

None of these

Question # 5 of 10 (Start time: 06:43:39 PM) Total Marks: 1

In binomial distribution, the experiment is repeated for

Select correct option:

Variable number of times

Fixed number of times

Many number of times

None of these

Question # 6 of 10 (Start time: 06:44:06 PM) Total Marks: 1

The probability of "success" p is equals to

Select correct option:

q

$1-q$

$1+q$

None of these

Question # 7 of 10 (Start time: 06:45:01 PM) Total Marks: 1

The successive negative binomial trials are assumed to be

Select correct option:

Independent

Dependent

May be independent

None of these

Question # 8 of 10 (Start time: 06:46:02 PM) Total Marks: 1

Probability of occurrence of an event A = $\frac{X}{n}$ (Where X=Number of outcomes favorable to event A and n=total number of outcomes)

Select correct option:

nX

n/X

X/n

None of these

Question # 9 of 10 (Start time: 06:46:58 PM) Total Marks: 1

Which of the following formula is correct?

Select correct option:

=BINOMDIST(7, 4, 0.5, FALSE)

=BINOMDIST(-4, 7, 1.5, FALSE)

=BINOMDIST(4, -7, 1.5, FALSE)

=BINOMDIST(4, 7, 0.5, FALSE)

Question # 10 of 10 (Start time: 06:47:55 PM) Total Marks: 1

The Excel command “=BINOMDIST(4, 10, 2, FALSE)” gives an error of #NUM.

The reason is that the probability of successes is 2 which is wrong. The probability of successes cannot be greater than 1.

Select correct option:

True

False

Probability of occurrence of an event A = $\frac{X}{n}$ (Where X=Number of outcomes favorable to event A and n=total number of outcomes)

Select correct option:

nX

n/X

X/n

None of these

What is probability of r or more successes when m=7;r=9

Select correct option:

0.2709

0.184

0.5

None of these

The probability of successes can be less than 0 or greater than 1.

Select correct option:

True

False

In Binomial distribution, the outcome on one trial the outcome on other trials.

Select correct option:

Does not affect

Does affect

May affect

None of these

This formula “=BINOMDIST(3, 6, 0.5, TRUE)” is for Cumulative Binomial distribution.

Select correct option:

True

False

Which of the following can be best fit for the prediction of the vehicles arriving at their destination in 2 hours?

Select correct option:

Binomial Distribution

Normal Distribution

Poisson Distribution

Chi-Square Distribution

Suppose, you have a bag in which there are one green and one red ball. You take out a ball repeatedly and put it back into the bag, and you count the number of times the green ball comes out. You continue to take out the ball until the green ball comes out 3 times. This experiment shows the case of Negative Binomial Distribution.

Select correct option:

True

False

The Excel command CRITBINOM(trials,p,alpha) where n is the number of trials, p is the probability of success in trials, and alpha is the criterion value. If trials < 0 , then CRITBINOM returns #NUM! error.

Select correct option:

True

Flase

In the excel formulation for Poisson distribution i-e "POISSON(x,mean,cumulative)", if X<0 then it returns to

Select correct option:

TRUE

FALSE

#VALUE!

#NUM!

Probability of sure event is -----

Select correct option:

0

1

-1

None of these

Question # 3 of 10 (Start time: 07:05:08 PM) Total Marks: 1

Probability of sure event is -----

Select correct option:

0

1

-1

None of these

Question # 4 of 10 (Start time: 07:05:35 PM) Total Marks: 1

In binomial distribution, each observation represents

Select correct option:

One of two outcomes ("success" or "failure")

Only one outcome ("success")

Only one outcome ("failure")

None of these

Question # 5 of 10 (Start time: 07:06:16 PM) Total Marks: 1

A pair of fair die is rolled. What will be the probability of "Sum of dots on upper sides is odd number?

Select correct option:

18/36

15/36

10/36

None of these

Question # 6 of 10 (Start time: 07:06:49 PM) Total Marks: 1

The Excel command CRITBINOM(trials,p,alpha) where n is the number of trials, p is the probability of success in trials, and alpha is the criterion value. If trials < 0 , then CRITBINOM returns #NUM! error.

Select correct option:

True

Flase

Question # 7 of 10 (Start time: 07:07:14 PM) Total Marks: 1

Suppose, you have a bag in which there are one green and one red ball. You take out a ball repeatedly and put it back into the bag, and you count the number of times the green ball comes out. You continue to take out the ball until the green ball comes out 3 times. This experiment shows the case of Negative Binomial Distribution.

Select correct option:

True

False

Question # 8 of 10 (Start time: 07:07:28 PM) Total Marks: 1

A fair die is rolled. What will be the probability of even numbers?

Select correct option:

1/6

2/6

3/6

None of these

Question # 9 of 10 (Start time: 07:08:31 PM) Total Marks: 1

An automatic production line breaks down every 2 hours.Special production requires uninterrupted operation for 8 hours.What is the probabilty that this can be achieved.

Select correct option:

0.312

0.021

0.0183

None of these.

Question # 10 of 10 (Start time: 07:10:01 PM) Total Marks: 1

Significant birth defects occur at a rate of about 4 per 1000 births in human populations. Which type of distribution can better deal with the above mentioned case?

Select correct option:

Binomial Distribution.

Normal Distribution.

Poisson Distribution.

None of the above.

Question # 1 of 10 (Start time: 07:12:03 PM) Total Marks: 1

In binomial distribution, the experiment is repeated for
Select correct option:

Variable number of times

Fixed number of times

Many number of times

None of these

Question # 1 of 10 (Start time: 07:12:03 PM) Total Marks: 1

In binomial distribution, the experiment is repeated for
Select correct option:

Variable number of times

Fixed number of times

Many number of times

None of these

Question # 3 of 10 (Start time: 07:13:44 PM) Total Marks: 1

Which of the following can be best fit for the prediction of the vehicles arriving at their destination in 2 hours?
Select correct option:

Binomial Distribution

Normal Distribution

Poisson Distribution

Chi-Square Distribution

Question # 4 of 10 (Start time: 07:14:31 PM) Total Marks: 1

The successive negative binomial trials are assumed to be
Select correct option:

Independent

Dependent

May be independent

None of these

Question # 5 of 10 (Start time: 07:15:06 PM) Total Marks: 1

In Binomial distribution, the outcome on one trial the outcome on other trials.

Select correct option:

Does not affect

Does affect

May affect

None of these

Question # 6 of 10 (Start time: 07:15:38 PM) Total Marks: 1

In the excel formulation for Poisson distribution i-e "POISSON(x,mean,cumulative)", if $X < 0$ then it returns to

Select correct option:

TRUE

FALSE

#VALUE!

#NUM!

Question # 7 of 10 (Start time: 07:15:58 PM) Total Marks: 1

Formulation for Poisson distribution i-e "POISSON(x,mean,cumulative)" is truncated if 'x' is

Select correct option:

Non-integer

Integer

Integer but expected numeric value

Integer but non-expected value

Question # 8 of 10 (Start time: 07:16:31 PM) Total Marks: 1

A lottery has Rs.100 payout on average 20 turns.What will be expected win per turn.

Select correct option:

Rs.5

Rs. -5

Rs.3

None of these.

Question # 9 of 10 (Start time: 07:17:11 PM) Total Marks: 1

If the mean & standard deviation on an examination are '74' & '12' respectively and the grades 'x=74' then the scores (z) in standard units of students receiving are

Select correct option:

0.8378
-0.8378
1
0

Question # 10 of 10 (Start time: 07:18:18 PM) Total Marks: 1

What is not a characteristic of poisson distribution.
Select correct option:

Either/or situation.
mean number of successes per unit,m,known and fixed.
p,chance unknown and large
None of these.

Question # 1 of 10 (Start time: 07:20:58 PM) Total Marks: 1

An automatic packing machine produces on an average one in 100 underweight bags.What is the probability that 500 bags contain less than three underweight bags?

Select correct option:

0.018
0.1247
0.312
None of these.

Question # 2 of 10 (Start time: 07:22:23 PM) Total Marks: 1

During a typical football game, a coach can expect 3.2 injuries. Find the probability that the team will have at most 1 injury in this game.We are interested in

Select correct option:

$P(X=1)$
 $P(X<1)$
 $P(X=1)$
 $P(X=0) + P(X=1)$

Question # 3 of 10 (Start time: 07:23:47 PM) Total Marks: 1

The probability of "success" p is equals to

Select correct option:

q
1-q
1+q
None of these

Question # 4 of 10 (Start time: 07:24:45 PM) Total Marks: 1

Which of the following can be best fit for the prediction of the vehicles arriving at

their destination in 2 hours?

Select correct option:

Binomial Distribution

Normal Distribution

Poisson Distribution

Chi-Square Distribution

Question # 5 of 10 (Start time: 07:25:14 PM) Total Marks: 1

Probability of occurrence of an event A = $\frac{X}{n}$ (Where X=Number of outcomes favorable to event A and n=total number of outcomes)

Select correct option:

nX

n/X

X/n

None of these

Question # 6 of 10 (Start time: 07:25:34 PM) Total Marks: 1

If the mean & standard deviation on an examination are '74' & '12' respectively and the standard scores is 'z= -1' then the grade 'x' is

Select correct option:

-62

889

887

62

Question # 7 of 10 (Start time: 07:26:49 PM) Total Marks: 1

In Binomial distribution, the outcome on one trial the outcome on other trials.

Select correct option:

Does not affect

Does affect

May affect

None of these

Question # 8 of 10 (Start time: 07:27:07 PM) Total Marks: 1

The marketing manager of a company receives average 10 complaint calls during a week .Suppose that the number of calls during a week follows the Poisson distribution. The probability that she gets five such calls in one day is:

Select correct option:

$(10^5 e^{-10})/5!$

$1-(10^5 e^{-10})/5!$

$10C5 * 0.55 * 0.55$

$1 - 10C5 * 0.55 * 0.55$

Question # 9 of 10 (Start time: 07:28:02 PM) Total Marks: 1

Poisson probability density function has ___ parameter(s)

Select correct option:

0

1

2

None of these.

Question # 10 of 10 (Start time: 07:28:51 PM) Total Marks: 1

The Excel command “=BINOMDIST(4, 10, 2, FALSE)” gives an error of #NUM.

The reason is that the probability of successes is 2 which is wrong. The probability of successes cannot be greater than 1.

Select correct option:

True

False

Question # 1 of 10 (Start time: 07:36:42 PM) Total Marks: 1

In binomial distribution, each observation represents

Select correct option:

One of two outcomes ("success" or "failure")

Only one outcome ("success")

Only one outcome ("failure")

None of these

Question # 2 of 10 (Start time: 07:38:02 PM) Total Marks: 1

Poisson probability density function has ___ parameter(s)

Select correct option:

0

1

2

None of these.

Question # 3 of 10 (Start time: 07:38:52 PM) Total Marks: 1

The Excel command CRITBINOM(trials,p,alpha) where n is the number of trials, p is the probability of success in trials, and alpha is the criterion value. If trials < 0 , then CRITBINOM returns #NUM! error.

Select correct option:

True

Flase

Question # 4 of 10 (Start time: 07:39:21 PM) Total Marks: 1

In the normal distribution curve, peak of the bell-shaped curve gives

Select correct option:

Standard deviation

Variance

Mean

Frequency

Question # 5 of 10 (Start time: 07:40:02 PM) Total Marks: 1

12% of the electric bulbs manufactured by a company are defective. What is the probability that a bulb selected will not be defective?

Select correct option:

90%

88%

80%

85%

Question # 6 of 10 (Start time: 07:40:31 PM) Total Marks: 1

A lottery has Rs.100 payout on average 20 turns.What will be expected win per turn.

Select correct option:

Rs.5

Rs. -5

Rs.3

None of these.

Question # 7 of 10 (Start time: 07:41:14 PM) Total Marks: 1

In the excel formulation for Poisson distribution i-e

"POISSON(x,mean,cumulative)", 'x' is the

Select correct option:

Expected numeric value

Logical value that determines the form of the probability distribution returned

Number of events over a specific time

Discrete non-random variable

Question # 8 of 10 (Start time: 07:41:45 PM) Total Marks: 1

Suppose, you have a bag in which there are one green and one red ball. You take out a ball repeatedly and put it back into the bag, and you count the number of times the green ball comes out. You continue to take out the ball until the green ball comes out 3 times. This experiment shows the case of Negative Binomial Distribution.

Select correct option:

True

False

Question # 9 of 10 (Start time: 07:42:30 PM) Total Marks: 1

What is the probability of at least one head in the experiment of tossing a coin three times?

Select correct option:

1/7

7/8

1/8

8/7

Question # 10 of 10 (Start time: 07:43:00 PM) Total Marks: 1

The probability of successes can be less than 0 or greater than 1.

Select correct option:

True
False

- - - - -

Probability distribution is a set of possible values that a random variable can assume and their associated probabilities of occurrence.
Select correct option:

True
False

In standard normal distribution;
Select correct option:

Mean = 1, Standard Deviation = 0
Mean = 0, Standard Deviation = 1
Mean = 1, Standard Deviation = 1
Mean = 0, Standard Deviation = 0

For normal distribution the probability of an event is zero if and only if value of 'z'
Select correct option:

Greater than 3
Less than 3
Equal to 3
None of these

For mutually exclusive events A and B
Select correct option:

Probability of A intersection B > 0
Probability of A intersection B < 0
Probability of A intersection B = 0
None of these

Which of the following formula is correct?
Select correct option:

=BINOMDIST(7, 4, 0.5, FALSE)
=BINOMDIST(-4, 7, 1.5, FALSE)
=BINOMDIST(4, -7, 1.5, FALSE)
=BINOMDIST(4, 7, 0.5, FALSE)

Probability of sure event is -----
Select correct option:

0
1

-1

None of these

In the excel formulation for Poisson distribution i-e
"POISSON(x,mean,cumulative)", 'x' is the
Select correct option:

Expected numeric value

Logical value that determines the form of the probability distribution returned

Number of events over a specific time

Discrete non-random variable

The probability of successes can be less than 0 or greater than 1.

Select correct option:

True

False

A pair of fair die is rolled. What will be the probability of "Sum of dots on upper
sides is odd number?

Select correct option:

18/36

15/36

10/36

None of these

An automatic production line breaks down every 2 hours.Special production
requires uninterrupted operation for 8 hours.What is the probability that this can
be achieved.

Select correct option:

0.312

0.021

0.0183

None of these.

Question # 1 of 10 (Start time: 08:27:06 PM) Total Marks: 1

Which of the following formula is correct?

Select correct option:

=BINOMDIST(7, 4, 0.5, FALSE)

=BINOMDIST(-4, 7, 1.5, FALSE)

=BINOMDIST(4, -7, 1.5, FALSE)

=BINOMDIST(4, 7, 0.5, FALSE)

Question # 2 of 10 (Start time: 08:27:39 PM) Total Marks: 1
The successive binomial trials are assumed to be
Select correct option:

Independent

Dependent

May be independent

None of these

Question # 3 of 10 (Start time: 08:28:27 PM) Total Marks: 1
For the Poisson distribution,
Select correct option:

$E(X) = \mu$, $\text{Var}(X) = \mu$

$E(X) = \mu$, $\text{Var}(X) = -\mu$

$E(X) = -\mu$, $\text{Var}(X) = \mu$

$E(X) = 0$, $\text{Var}(X) = 0$

Question # 4 of 10 (Start time: 08:28:48 PM) Total Marks: 1
Probability distribution is a set of possible values that a random variable can assume and their associated probabilities of occurrence.
Select correct option:

True

False

Question # 5 of 10 (Start time: 08:29:20 PM) Total Marks: 1
The marketing manager of a company receives average 10 complaint calls during a week .Suppose that the number of calls during a week follows the Poisson distribution. The probability that she gets five such calls in one day is:
Select correct option:

$(10^5 e^{-10})/5!$

$1 - (10^5 e^{-10})/5!$

$10C5 * 0.55 * 0.55$

$1 - 10C5 * 0.55 * 0.55$

Question # 6 of 10 (Start time: 08:30:29 PM) Total Marks: 1
During a typical football game, a coach can expect 3.2 injuries. Find the probability that the team will have at most 1 injury in this game.We are interested in
Select correct option:

$P(X=1)$

$P(X < 1)$

$P(X=1)$

$P(X=0) + P(X=1)$

Question # 7 of 10 (Start time: 08:31:03 PM) Total Marks: 1
For mutually exclusive events
Select correct option:

$P(A \cup B) = P(A)P(B)$
 $P(A \cup B) = P(A)/P(B)$
 $P(A \cup B) = P(A) - P(B)$
 $P(A \cup B) = P(A) + P(B)$

Question # 8 of 10 (Start time: 08:32:27 PM) Total Marks: 1
In binomial distribution, the probability of "success" p
Select correct option:

Vary for each outcome

Remains constant for different outcomes

Remains the same for each outcome

None of these

Question # 9 of 10 (Start time: 08:33:50 PM) Total Marks: 1
In the excel formulation for Poisson distribution i-e
"POISSON(x,mean,cumulative)", if $X < 0$ then it returns to
Select correct option:

TRUE

FALSE

#VALUE!

#NUM!

Question # 10 of 10 (Start time: 08:34:31 PM) Total Marks: 1
A fair die is rolled. What will be the probability of even numbers?
Select correct option:

1/6

2/6

3/6

None of these

Question # 1 of 10 (Start time: 08:40:15 PM) Total Marks: 1
If the mean & standard deviation on an examination are '74' & '12' respectively
and the standard scores is ' $z = -1$ ' then the grade ' x ' is
Select correct option:

-62

889

887

62

Question # 2 of 10 (Start time: 08:41:47 PM) Total Marks: 1
What is not a characteristic of poisson distribution.
Select correct option:

Either/or situation.
mean number of successes per unit, m , known and fixed.
 p , chance unknown and large
None of these.

Question # 3 of 10 (Start time: 08:42:34 PM) Total Marks: 1
Which of the following is applicable to a Poisson Distribution?
Select correct option:

It is used to compute the probability of rare events.
Every event is independent of every other event.
The range for the number of events that could occur is 0,1,2,3, ...

all of given choices.

Question # 4 of 10 (Start time: 08:43:34 PM) Total Marks: 1
The successive binomial trials are assumed to be
Select correct option:

Independent
Dependent
May be independent
None of these

Question # 5 of 10 (Start time: 08:44:21 PM) Total Marks: 1
Which of the following can be best fit for the prediction of the vehicles arriving at their destination in 2 hours?
Select correct option:

Binomial Distribution
Normal Distribution
Poisson Distribution
Chi-Square Distribution

Question # 6 of 10 (Start time: 08:45:00 PM) Total Marks: 1
What is the range of probability of any event?
Select correct option:

-1 to 0
-1 to 1
0 to 1
None of these

Question # 9 of 10 (Start time: 08:47:56 PM) Total Marks: 1
A lottery has Rs.100 payout on average 20 turns. What will be expected win per turn.
Select correct option:

Rs.5
Rs. -5
Rs.3
None of these.

Question # 10 of 10 (Start time: 08:48:42 PM) Total Marks: 1

The marketing manager of a company receives average 10 complaint calls during a week. Suppose that the number of calls during a week follows the Poisson distribution. The probability that she gets five such calls in one day is:

Select correct option:

$(10^5 e^{-10})/5!$

$1 - (10^5 e^{-10})/5!$

$10C5 * 0.55 * 0.55$

$1 - 10C5 * 0.55 * 0.55$

A coin is tossed twice. What is the probability of getting head on first toss and tail on second toss?

Select correct option:

$\frac{1}{2}$

$\frac{1}{3}$

$\frac{1}{4}$

None of these

A lottery has Rs.100 payout on average 20 turns. What will be expected win per turn.

Select correct option:

Rs.5

Rs. -5

Rs.3

None of these.

$20/100=5$

In binomial distribution, each observation represents

Select correct option:

One of two outcomes ("success" or "failure")

Only one outcome ("success")

Only one outcome ("failure")

None of these

An automatic packing machine produces on an average one in 100 underweight bags. What is the probability that 500 bags contain less than three underweight bags?

Select correct option:

0.018

0.1247

0.312

None of these

The correct formula for OR rule is

Select correct option:

$P(A \text{ or } B) = P(A) + P(B)$

$P(A \text{ or } B) = P(A) - P(B)$

$P(A \text{ or } B) = P(A) + P(B)$

None of these

In Binomial distribution, the outcome on one trial the outcome on other trials.

Select correct option:

Does not affect

Does affect

May affect

None of these

In the excel formulation for Poisson distribution i-e "POISSON(x,mean,cumulative)", 'x' is the

Select correct option:

Expected numeric value

Logical value that determines the form of the probability distribution returned

Number of events over a specific time

Discrete non-random variable

The successive binomial trials are assumed to be

Select correct option:

Independent

Dependent

May be independent

None of these

Probability distribution is a set of possible values that a random variable can assume and their associated probabilities of occurrence.

Select correct option:

True

False

The Excel command " =BINOMDIST(4, 10, 2, FALSE)" gives an error of #NUM.

The reason is that the probability of successes is 2 which is wrong. The probability of successes cannot be greater than 1.

Select correct option:

True

False

The correct formula for AND rule is

Select correct option:

$P(A \text{ and } B) = P(A) P(B)$

$P(A \text{ and } B) = P(A) - P(B)$

$P(A \text{ and } B) = P(A) + P(B)$

None of these